Greatest Common Divisor and Division Thm.

- **Def.** If m and n are nonzero integers, then GCD(m,n) is the largest element of $Div(m) \cap Div(n)$
- **Ex.** GCD(5,15)=5
- **Thm.** (The Division Thm.) $\forall n, d > 0, \exists!\, q, r\ so\ that\ n = d \cdot q + r$ and $0 \leq r < d$
  - Note: we call q the quotient, d the divisor, and r the remainder
- **Pf. 1** (Existence, nonconstructive)
  - Consider the set $A = \{n - d \cdot q \mid q \in \mathbb{Z}\}$ "The remainder set"
  - Since $d \neq 0, q$ can be positive or negative, A must have a non-negative member
  - Therefore, A must have a *least nonnegative* member, called r
  - Now $r < d$ (why?)
    - because otherwise by subtraction, $0 \leq r - d < r$ would be a non-negative element of A, but this contradicts the minimality of r
  - So Let $q \in \mathbb{Z}$ witness that $r \in A$, then notice $r = n - d \cdot q \Rightarrow n = d \cdot q + r$

Pf. 2: Uniqueness of division theorem

- Now need uniqueness!
- Suppose, for contradiction, that for fixed (but arbitrary) $n, d > 0\ \exists q_1, q_2, r_1, r_2$ so that:
  - $n = q_1 \cdot d + r_1$
  - $n = q_2 \cdot d + r_2$
- Then, by subtracting both sides:
  - $0 = q_1 \cdot d + r_1 - q_2 \cdot d - r_2 = \mathrm{d}(\mathrm{q}_1 - \mathrm{q}_2) + \mathrm{r}_1 - \mathrm{r}_2$ (*)
- But now we claim: $d|(r_1 - r_2)$ (**)
  - That's because d|LHS and $d|d(q_1 - q_2)$ so must be the case $d|(r_1 - r_2)$
  - This is a bit subtle! Recall that in general $d|(a + b)$ doesn't mean $d|a\ and\ d|b$.
- Claim: Since $0 \leq r_1 - r_2 < d,\ (**)\ \Rightarrow r_1 - r_2 = 0 \Rightarrow r_1 = r_2$
- So now $\mathrm{d} \cdot (q_1 - q_2) = 0$ by (*), but $d > 0$, so must be that $q_1 - q_2 = 0$
- So $q_1 = q_2$. QED

Division theorem corollary

- So division theorem tells us that $\forall n, d > 0, n = q \cdot d + r$ where $r < d$, and this is unique
- This motivates the definitions:
  - $n\ \boldsymbol{div}\ d = q$ and
  - $n\ \boldsymbol{mod}\ d = r$
  - as the q and r whose uniqueness is guaranteed by the division theorem.
- **Thm.** Suppose $d = \gcd(m, n)$. Then there exists a, b so that $d = a \cdot m + b \cdot n$
  - i.e., the gcd of m and n can be written as a linear combination of m and n

Division theorem corollary (proof)

- **Thm.** Suppose $d = \gcd(a, b)$. Then there exist integers x, y so that $d = a \cdot x + b \cdot y$
- (These integers x,y are not necessarily unique!)
- **Pf.**
  - First, if a=b=0 this is trivially true, so wlog assume at least one of a, $b \neq 0$.
  - Consider set $S = \{a \cdot x + b \cdot y > 0\colon\ x, y \in \mathbb{Z}\}$
  - Because S is a nonempty set of positive integers…
    - It has a minimum element $d = a \cdot s + b \cdot t$
  - Claim that $d = \gcd(a, b)$.
  - First will show that d is a common divisor of a,b. WLOG show this for a (same for b)
    - Divide a by d, by division thm.: $a = dq + r$ with $0 \leq r < d$.
    - Claim: The remainder, r, is in $S \cup \{0\}$
      - because: $r = a - dq = a - (as + bt)q = a(1 - qs) - b(qt)$ (by substitution and algebra)
    - But recall d is the smallest positive integer in S *and* r being a remainder, must be less than d. So r=0.
    - But this means that $d|a$ (and by same token $d|b$).
  - Now we'll show d is **greatest** common divisor of a and b
    - Let $c \in div(a) \cap div(b)$. Then there exists u, v so that $a = cu$ and $b = cv$
    - Then $d = as + bt$ (from above) = $cus + cvt$ (by substitution) $= c(us + vt)$
    - But this means that c divides d! Then $c \leq d$. QED

Euclid's algorithm (preliminaries)

- Euclid's algorithm for computing GCD(a,b).
  - One of the first algorithms!
  - What is an algorithm?
    - Intuitively, a procedure for solving a problem without going through all possible inputs
- **Starting Thm**: If $d = \gcd(a, b)$, $b \neq 0$ and $r = a\ \boldsymbol{mod}\ b$ then $d = \gcd(b, r)$.
  - i.e., if d is the gcd of a and b, then d is also the gcd of b and the remainder of a divided by b
- **Pf.** We'll show $Div(a) \cap Div(b) = Div(b) \cap Div(r)$
  - That is, that the common divisors of both a and b are the same as the common divisors of b and r
  - First, let $q = a\ \boldsymbol{div}\ b$, so say $a = q \cdot b + r$, by definition
  - Now let $z \in Div(a) \cap Div(b)$
    - Then $z|a$ and $z|b \cdot q \Rightarrow z|(a - bq) = r$
    - So $z \in Div(r)$
  - Now let $z' \in Div(b) \cap Div(r)$
    - Then $z'|(q \cdot b + r) = a$, so $z' \in Div(a)$
- Now, recall $d = \gcd(a, b) = \max(Div(a) \cap Div(b))$ so must also be $\gcd(\mathrm{b}, \mathrm{r}) = \max(Div(b) \cap Div(r))$ QED.

Euclid’s algorithm

- The **starting thm.** on the last slide gives way immediately to an **algorithm** for computing gcd(a,b)
- Consider, we’ve just proven:

$$\begin{aligned} gcd(a,b) &= a,\ if\ b = 0 \\ &= gcd(b, a \mod b)\ otherwise \end{aligned}$$

- So can compute by continually swapping the “b” argument for the remainder: a mod b, until we get 0!
- **Ex:** $\gcd(15,9) = \gcd(9,15\ mod\ 9 = 6) = \gcd(6,3) = \gcd(3,0) = 3$